🔑 **Answer Key — Teacher Copy · Lesson 7-7**

MSTAR PRACTICE WORKSHEET

Lesson 7-7: Represent Polygons on the Coordinate Plane

Grade 6 Mathematics · Standard 6.NOS.9

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: D

Vertices (-4.5, 3) and (5, 3) share y = 3. What is the distance between them?

A. 0.5 units

B. 1.5 units

C. 8 units

D. 9.5 units

They differ only in x: |5 − (−4.5)| = 5 + 4.5 = 9.5 units.

Question 2 SELECTED RESPONSE — CORRECT: C

Vertices (5, 3) and (5, -3.75) share x = 5. What is the distance between them?

A. 0.75 units

B. 5 units

C. 6.75 units

D. 8.75 units

They differ only in y: |3 − (−3.75)| = 3 + 3.75 = 6.75 units.

Question 3 SELECTED RESPONSE — CORRECT: C

What is the perimeter of the rectangle with side lengths 9.5 and 6.75 units?

A 16.25 units

B 26 units

C 32.5 units

D 64.125 units

Perimeter = 2(9.5) + 2(6.75) = 19 + 13.5 = 32.5 units.

Question 4 SELECTED RESPONSE — CORRECT: D

Why are two of the terms multiplied by 2 in the perimeter calculation?

A Because perimeter always doubles the area

B Because there are two coordinates in each ordered pair

C It is a rounding rule

D A rectangle's opposite sides are equal, so each length appears on two sides

The two 9.5 sides and the two 6.75 sides each come in pairs — that is what the 2 counts.

Question 5 SELECTED RESPONSE — CORRECT: D

To draw a polygon from a list of vertices, you:

A Plot only the first vertex

B Connect the points in any random order

C Add all the coordinates together

D Plot each ordered pair, then connect them in order with segments

Graph the points on the plane and connect them in order — the closed figure that appears is the polygon.

Question 6 SELECTED RESPONSE — CORRECT: D

The park's marked vertices are 3 units apart and each unit is 100 feet. What is the park's width?

A. 3 feet

B. 30 feet

C. 103 feet

D. 300 feet

3 units × 100 feet per unit = 300 feet.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

A rectangle has vertices at (1, 1), (7, 1), (7, 5), and (1, 5). What is its perimeter?

A. 10 units

B. 12 units

C. 20 units

D. 24 units

The horizontal sides are $|7 - 1| = 6$ units and the vertical sides are $|5 - 1| = 4$ units. Perimeter = $2(6) + 2(4) = 20$ units.

- **A:** That adds one of each side. Each length appears twice around a rectangle.
- **B:** That doubles the horizontal side only. The two vertical sides of 4 units each still count.
- **D:** That is the AREA, 6 times 4, in square units. Perimeter adds the four side lengths.

Part B — correct answer: B

How can the side lengths be found from the coordinates alone?

A. Count every grid square inside the rectangle.

B. Each side is horizontal or vertical, so its length is the difference of the coordinates that change.

C. Add all eight coordinate values together.

D. Multiply the x-coordinates by the y-coordinates.

Every side of this rectangle runs along a grid line, so one coordinate stays fixed and the length is the absolute difference of the other.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about polygons on the coordinate plane:

- **A** Vertices are connected in the order given to form the polygon.
- **B** A horizontal side's length is the absolute difference of the x-coordinates.
- **C** A vertical side's length is the absolute difference of the y-coordinates.
- **D** The length of a slanted side can be found by subtracting coordinates.
- **E** The area of a rectangle on the grid is the product of its side lengths.

A, B, C, and E are true. D is false — a slanted side changes in BOTH directions, so subtracting one pair of coordinates does not give its length.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

The park design used width from coordinates, then length from the area, then the scale to place the missing vertices. Design your own version: pick an area and a scale, mark two vertices, and show where the other two go.

Model answer: My park has area 60,000 sq ft with each unit = 100 ft. I mark vertices at (1, 5) and (3, 5) — 2 units apart, so width 200 ft. Length = 60,000 ÷ 200 = 300 ft = 3 units, so the other vertices go at (1, 2) and (3, 2), 3 units below.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.